WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

Dataset Basics and Dirty Data



Rows and columns

ID	Major	Year	Sleep (hrs)
101	Biology	Freshman	7
102	Nursing	Junior	6
103	History	Sophomore	8
104	Biology	Senior	5

Row = observation

One student, one sale, one day.

Column = variable

One kind of fact about every row.

First question: what is one row?

Dataset	One row is	Example columns
Student survey	One student	Major, year, sleep hours
Campus café	One sale	Item, price, time
Melt pool (Assignment 1)	One laser test	Laser power, scan speed, defect type

Ask “what is one row?” before anything else.

A data set is not a data source

Data source

Where the data comes from: a website, a sensor, a survey, a database.

Data set

The table you load and work on: rows and columns, often one file.

Example: source = a city open-data website. Data set = the bike-share file you downloaded from it.

One source can give many data sets. One data set can mix several sources.

Three types of data

Numeric

Numbers you can add and average.

Examples

Age, price, temperature

Labels

Names of groups, with no order.

Examples

Major, city, color

Ordered labels

Groups with a natural order.

Examples

Small, medium, large

Quick test: can I average it? Numeric yes. Labels no. Ordered labels, not really.

Numeric data

What it is

Numbers you can do math with.

What you can do

Mean, median, min, max, range, histogram.

Watch out

ID numbers and zip codes look numeric but are labels.

Column	Example
Age	19
Price ($)	4.99
Temperature (°C)	21.5
Zip code	45435

Zip code looks like a number but is a label.

Labels (categorical data)

What it is

Names of groups. No order, no math.

What you can do

Count, percent, bar chart, compare groups.

Watch out

Typos split one group into two.

Column	Example
Major	History
City	Dayton
Color	Green
Student ID	101

Labels are counted, not averaged.

Ordered labels (ordinal data)

What it is

Groups with a natural order.

What you can do

Sort, count, compare which is higher.

Watch out

Gaps are not equal. Be careful averaging.

Column	Example order
Year	Freshman to Senior
T-shirt size	S, M, L, XL
Star rating	1 to 5 stars
Agreement	Disagree to Agree

The order matters, the gaps do not.

Nominal, ordinal, interval, ratio

Our word	Statistics word	Example	What you can do
Labels	Nominal	Platform: PC, Console, Mobile	Count them. Same or different. No order.
Ordered labels	Ordinal	Skill Tier: Bronze to Diamond	Put in order. Gaps between steps are unknown.
Numbers with a true zero	Ratio	Hours played (0 = none)	Add, subtract, divide. 20 is twice 10.
Numbers with no true zero	Interval	Temperature in °C	Add, subtract. 20 °C is not twice 10 °C.

Sort the columns

Column	Example value	Type
Age	19	Numeric
Zip code	45435	Label
T-shirt size	M	Ordered label
Major	History	Label
Temperature	21.5	Numeric
Star rating	4 stars	Ordered label
Student ID	101	Label

What is dirty data?

Missing values

A blank where a value should be.

Duplicates

The same row appears twice.

Typos

“Biolgy” instead of “Biology”.

Outliers

A value far from the rest.

Junk data

Rows that are not real data.

Even good sources have dirty data.

Missing values

ID	Major	Sleep (hrs)	Commute (min)
101	Biology	7	15
102	Nursing		20
103	History	8	
104	Biology	5	25

Two blanks. Zero is not blank.

Fill it

Put a sensible number in (median).

Keep as is

Leave the blank if it does not matter.

Remove the row

If the row is not usable.

Filling a blank with the median

1 Sort the known numbers.

2 Find the middle. With two middle numbers, average them.

3 Type that number into the blank.

Known: 10, 15, 15, 20, 25, 180

Middle two: 15 and 20

Median = 17.5

Mean = 44.2

The 180 drags it up.

Use the median. One extreme value does not move it. Labels: use the most common one.

Values go missing for different reasons

Not recorded	A sensor failed, or nobody typed it in.
Skipped	A person left the question blank.
Not applicable	Blank Defect Type means no defect.
Hidden	High earners leave salary empty.
Lost	A file error dropped the value.

Missing data follows one of three patterns

Short	Full name	Plain meaning	Example
MCAR	Missing completely at random	Chance only. Nothing explains it.	A sensor loses power for a moment.
MAR	Missing at random	Depends on another column we can see.	Mobile players more often have a blank ping. Platform is recorded.
MNAR	Missing not at random	Depends on the missing value itself.	High earners leave salary blank.

The pattern decides which fix is safe.

MCAR: missing completely at random

Blanks land anywhere. No pattern.

12	14	–	13	15	–	11	–	14	13

What it means

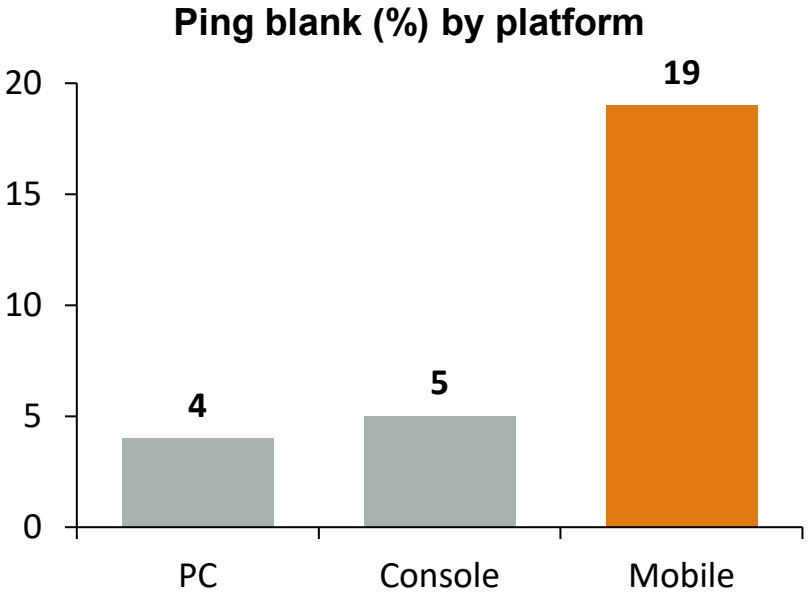
The blank has nothing to do with any data, seen or unseen. Example: a sensor loses power for a moment.

What to do

Delete the rows with blanks, or analyze as usual. Results stay unbiased.

Cost: fewer rows, so less certainty.

MAR: missing at random



What it means

The blank depends on a column we can see (Platform), not on the ping itself.

What to do

Use the other columns: model-based imputation, maximum likelihood, or multiple imputation (several filled-in copies, results combined).

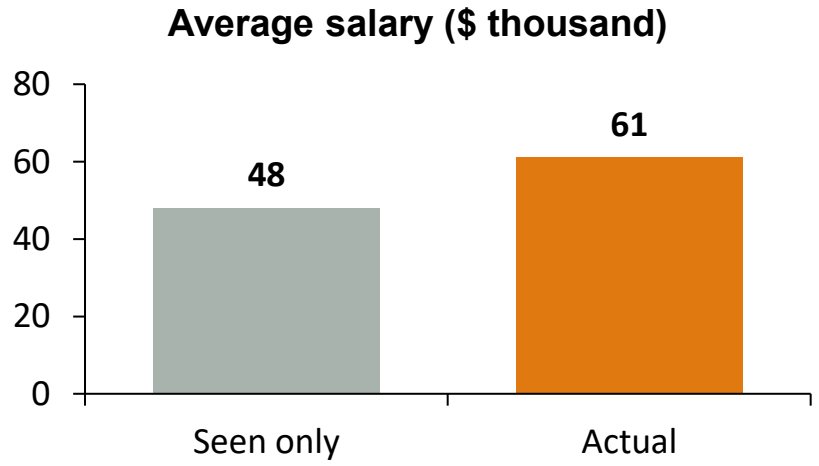
MNAR: missing not at random

What it means

The blank depends on the missing value itself. Example: high earners leave salary blank.

What to do

Model why it is missing, or test it: try several guesses for the hidden values. Does the answer change?



Ignoring it makes the group look poorer than it is.

You can test for a pattern in Orange

1 Make a Yes/No column: is Ping blank? (Formula widget)

2 Compare it with other columns (Box Plot or Distributions, split by it)

3 If the groups differ, it is not pure chance (MAR or MNAR)

Example

Ping blank: PC 4%, Console 5%, Mobile 19%.

It depends on Platform, so this is MAR.

Careful

Data cannot prove MNAR. Use what you know about how it was collected.

Each pattern has a safe fix

Pattern	Safe fix	Risk if ignored
MCAR	Delete the rows, or fill with the median.	Fewer rows, same answer.
MAR	Use other columns: model-based or multiple imputation.	A plain fill can hide the pattern.
MNAR	Model it, or run a sensitivity check.	Biased results.

In this class: few blanks and no pattern? Fill with the median and say so in your report.

Keep the row or remove it?

Situation	Decision
One blank in a minor column	Keep the row. Fill with the median.
Blank in the column you want to answer	Remove the row. It cannot teach us.
More than half the row is blank	Remove the row.
The blank means “none”	Keep the row. Write None.
Most of one column is blank	Consider dropping that column.

Duplicates

ID	Name	Major	Sleep
101	Ava	Biology	7
102	Ben	Nursing	6
102	Ben	Nursing	6
103	Chen	History	8
104	Ava	Nursing	7

Exact copy

Remove one copy.

Same ID, different values

Investigate. Keep the right one.

Same name, different ID

Probably a different person. Keep.

Typos and inconsistent labels

ID	Major
101	Biology
102	biology
103	Biolgy
104	BIO
105	Nursing

Before: 4 “groups”

Biology 1, biology 1, Biolgy 1, BIO 1

After: 1 group

Biology 4

Unsure what it meant? Make it blank.

Mixed units need one standard unit

Problem	Example	Fix
Mixed units	6.0 ft and 1.8 m in one column	Convert to meters: m = ft × 0.3048
Currency	$12 and €10 in one column	Pick one currency and multiply by a stated rate
Time zones	9:00 EST and 6:00 PST	Convert everything to one zone
Time units	90 minutes and 1.5 hours	Convert to one unit: hours = minutes ÷ 60

Add a converted column, keep the old

Height as typed	Unit	Height (m)
6.0	ft	1.83
1.80	m	1.80
72	in	1.83
5.5	ft	1.68

How

You need to know the unit of every row.

Then multiply: ft × 0.3048, in × 0.0254.

Keep the original column. Add the new one.

Outliers

Value	Can it be real?	Decision
Sleep 3 hours	Unusual, but possible	Keep. Flag it.
Age 190	Impossible	Treat as blank, then fill or remove.
Sleep 25 hours	Impossible (a day has 24)	Treat as blank, then fill or remove.
Commute 180 min	Unusual, could be real	Keep. It pulls the mean up.

Ask: can this value be real? Yes: keep. No: fix or blank.

Junk data

ID	Major	Year	Sleep
101	Biology	Freshman	7
test	asdf	???	999
-	-	-	-
0	0	0	0

Whole row junk

Remove the row.

One junk cell

Make it blank, then decide.

Decision cheat sheet

Problem	Example	Common decision
Missing value	Sleep is blank	Fill with median, or remove if row unusable
Duplicate	Same row twice	Remove one copy
Typo	“Biolgy”	Standardize the spelling
Outlier, could be real	Commute 180 min	Keep and flag
Outlier, impossible	Age 190	Treat as blank, fill or remove
Junk	“test, asdf, ???”	Remove the row

Try it: find the problems

#	Name	Major	Sleep (hrs)	Commute (min)
1	Ava	Biology	7	15
2	Ben	Nursing		20
3	Ben	Nursing		20
4	Chen	Hisotry	8	25
5	Dana	Biology	6	180
6	Eli	Nursing	31	10
7	test	asdf	???	999
8	Fay	History	8	

Answers

#	Problem	Decision
2	Sleep is blank	Fill with the median, 7.5
3	Duplicate of row 2	Remove one copy
4	Typo “Hisotry”	Change to History
5	Commute 180: unusual, could be real	Keep and flag
6	Sleep 31: impossible	Treat as blank, fill with 7.5
7	Junk row	Remove the whole row
8	Commute is blank	Fill with the median, 20

Write down what you changed

Good habits

Keep the original file.

Save a cleaned copy.

Log every change and why.

Row	Change	Why
4	Hisotry to History	Typo
7	Removed	Junk row
2	Sleep filled: 7.5	Blank, median

Example log. Anyone can check your decisions.

Excel to CSV to Orange

1 In Excel: File, Save As, CSV UTF-8 (Comma delimited)

2 Check: one header row, no merged cells, no notes above the table

3 In Orange: drag the File widget onto the canvas

4 Double-click it and browse to the CSV file

Why CSV?

Plain text. Opens everywhere. No hidden formatting.

Orange can read .xlsx too, but CSV is the course standard.

Check what Orange loaded

Check	What to look for	If it is wrong
Rows and columns	The counts match the file	Reload and check the file path
Column types	Numbers are numeric, labels are categorical	Change the type in the File widget
Column roles	ID column set to skip or meta, target set	Change the role in the File widget
Names	Clear, no odd symbols	Rename in Edit Domain